

Initiating Search

November 10, 2025, 4:44 PM

• Search:

CAS Lexicon Search:

Difluoromethylation - Preferred Concept

Search Tasks

Task	Result Type	View
Returned Reference Results from CAS Lexicon (562)	 References	View Results
Exported: Retrieved Related Reaction Results + Filters (312)	 Reactions	View Results
Filtered By:		
Yield:	90-100%, 80-89%, 70-79%	
Document Type:	Journal	

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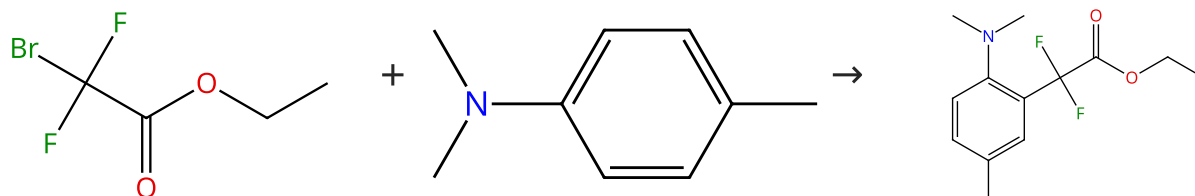
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Reactions (50)

[View in CAS SciFinder](#)

Scheme 1 (1 Reaction)

Steps: 1 Yield: 99%


 Suppliers (78)

 Suppliers (75)

31-614-CAS-38961850

Steps: 1 Yield: 99%

Photo-induced C(sp²)-H difluoroalkylation of anilines

1.1 Reagents: Sodium carbonate
Solvents: Acetonitrile; 24 h, rt

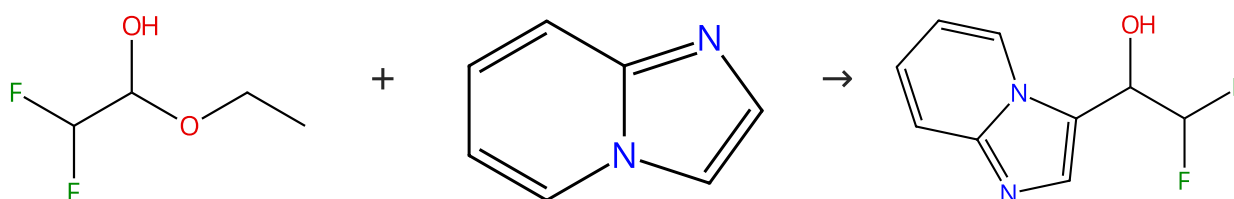
By: Zhu, Xing-Li; et al

Organic & Biomolecular Chemistry (2024), 22(4), 720-724.

Experimental Protocols

Scheme 2 (1 Reaction)

Steps: 1 Yield: 97%


 Suppliers (62)

 Suppliers (85)

 Suppliers (5)

31-614-CAS-38724260

Steps: 1 Yield: 97%

1,1,1,3,3,3-Hexafluoro-2-Propanol-Promoted Friedel-Crafts
Reaction: Metal-Free Synthesis of C3-Difluoromethyl Carbinol-
Containing Imidazo[1,2-a]pyridines at Room Temperature

1.1 Solvents: 1,1,1,3,3,3-Hexafluoro-2-propanol; 12 h, rt

By: Gao, Juanjuan; et al

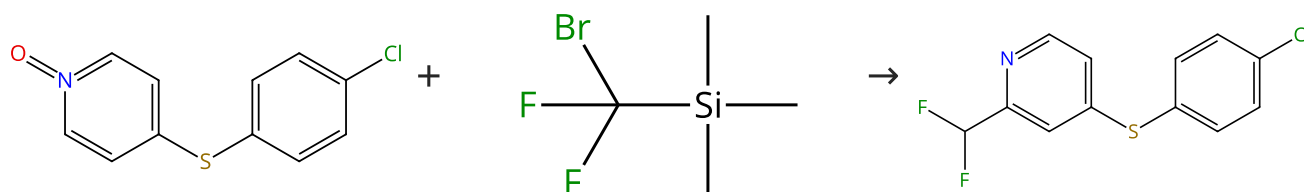
Molecules (2023), 28(22), 7522.

1.2 Solvents: Water; rt

Experimental Protocols

Scheme 3 (1 Reaction)

Steps: 1 Yield: 97%

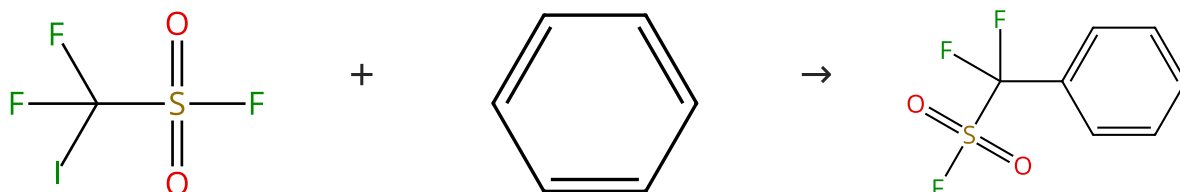

 Supplier (1)

 Suppliers (56)

31-614-CAS-24002265	Steps: 1 Yield: 97%	Synthesis of Difluoroalkylated Heteroarenes via Difluoro carbene
1.1 Reagents: Methyl triflate Solvents: Dichloromethane; rt; 1 h, rt; rt → -20 °C		By: Trifonov, Alexey L.; et al
1.2 Reagents: Triphenylphosphine Solvents: Acetonitrile; -30 °C; -30 °C → -42 °C		Organic Letters (2021), 23(17), 6977-6981.
1.3 Reagents: Hexamethylphosphoramide; -42 °C; 2 h, -40 °C; 1 h, -40 °C → -16 °C; 15 min, rt; rt → -20 °C		
1.4 Reagents: Triethylamine, Water; -20 °C; 12 h, rt		
Experimental Protocols		

Scheme 4 (1 Reaction)

Steps: 1 Yield: 96%



Supplier (1)

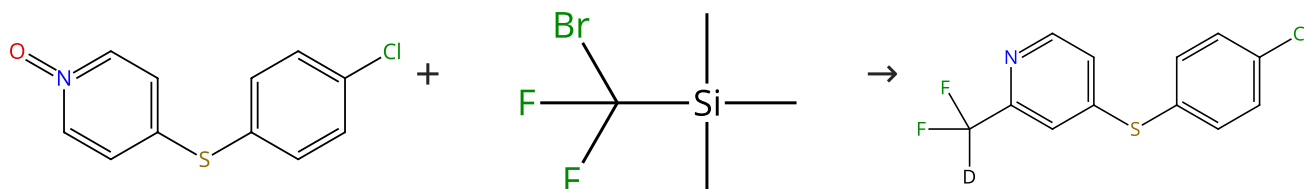
Suppliers (174)

Suppliers (4)

31-614-CAS-41437313	Steps: 1 Yield: 96%	Photoredox-Catalysis Fluorosulfonyldifluoromethylation of Unactivated Alkenes and (Hetero)arenes with ICF₂SO₂F
1.1 Reagents: Silver triflate Catalysts: Iridium(1+), (2,2'-bipyridine-κ ^N ¹ ,κ ^N ¹)bis[3,5-difluoro-2-[5-(trifluoromethyl)-2-pyridinyl-κ ^N]phenyl-κ ^C]-, (O ⁻ C-6-33)-, hexafluorophosphate(1-)(1:1) Solvents: Dimethyl carbonate; rt; 12 h, rt		By: He, Xu-Biao; et al
Experimental Protocols		Organic Letters (2024), 26(32), 6900-6904.

Scheme 5 (1 Reaction)

Steps: 1 Yield: 96%



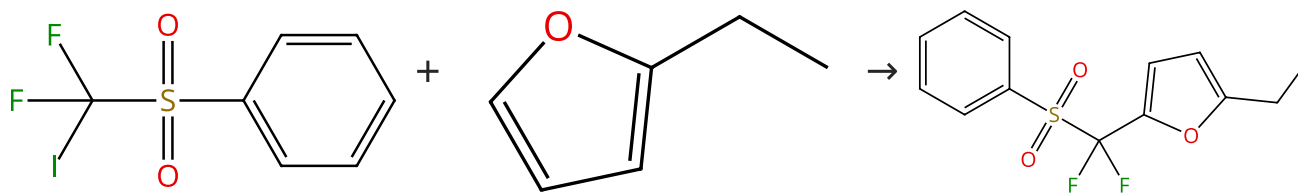
Supplier (1)

Suppliers (56)

31-614-CAS-24002259	Steps: 1 Yield: 96%	Synthesis of Difluoroalkylated Heteroarenes via Difluoro carbene
1.1 Reagents: Methyl triflate Solvents: Dichloromethane; rt; 1 h, rt; rt → -20 °C		By: Trifonov, Alexey L.; et al
1.2 Reagents: Triphenylphosphine Solvents: Acetonitrile; -30 °C; -30 °C → -42 °C		Organic Letters (2021), 23(17), 6977-6981.
1.3 Reagents: Hexamethylphosphoramide; -42 °C; 2 h, -40 °C; 1 h, -40 °C → -16 °C; 15 min, rt; rt → -20 °C		
1.4 Reagents: Triethylamine, Water- <i>d</i> ₂ ; -20 °C; 12 h, rt		
Experimental Protocols		

Scheme 6 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (51)

Suppliers (74)

31-614-CAS-44133246

Steps: 1 Yield: 96%

Pd-catalyzed C-H(phenylsulfonyl)difluoromethylation of electron-rich heteroarenes

1.1 **Reagents:** Cesium carbonate
Catalysts: Tris(dibenzylideneacetone)dipalladium, 1,1'-(9,9-Dimethyl-9H-xanthene-4,5-diyl)bis[1,1-diphenylphosphine]
Solvents: Chloroform; 10 min, rt; 24 h, 60 °C

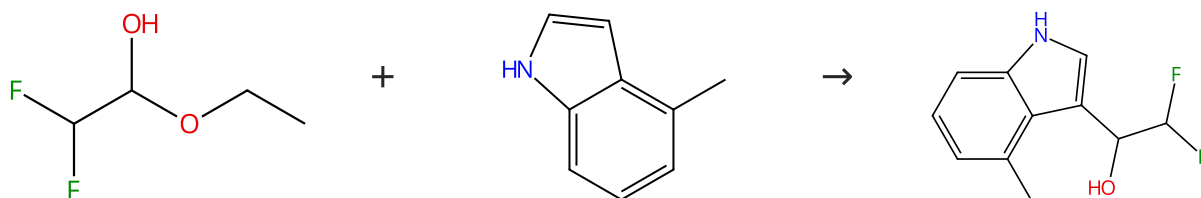
By: Liu, Zi-En; et al

Tetrahedron Letters (2025), 158, 155486.

Experimental Protocols

Scheme 7 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (62)

Suppliers (84)

31-614-CAS-36085901

Steps: 1 Yield: 96%

Synthesis of Difluoromethylated Carbinols via a HFIP-Promoted Hydroxydifluoromethylation of Aniline, Indole, and Pyrrole Derivatives with Difluoroacetaldehyde Ethyl Hemiacetal

1.1 **Reagents:** 1,1,1,3,3,3-Hexafluoro-2-propanol
Solvents: Dichloromethane; 48 h, rt

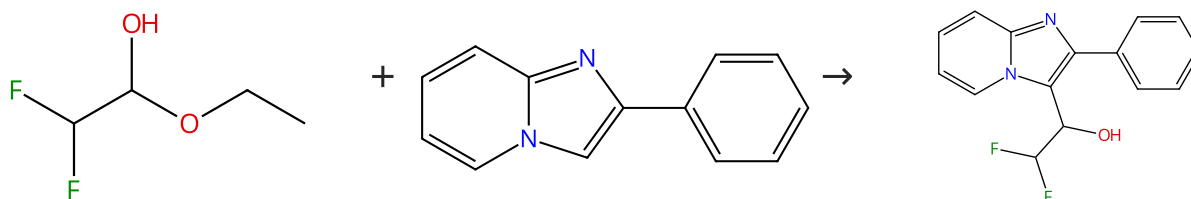
By: Yang, Jianguo; et al

Journal of Organic Chemistry (2023), 88(7), 4790-4798.

Experimental Protocols

Scheme 8 (1 Reaction)

Steps: 1 Yield: 95%



Suppliers (62)

Suppliers (80)

31-614-CAS-38724264

Steps: 1 Yield: 95%

1,1,1,3,3,3-Hexafluoro-2-Propanol-Promoted Friedel-Crafts Reaction: Metal-Free Synthesis of C3-Difluoromethyl Carbinol-Containing Imidazo[1,2-a]pyridines at Room Temperature

1.1 **Solvents:** 1,1,1,3,3,3-Hexafluoro-2-propanol; 12 h, rt1.2 **Solvents:** 1,1,1,3,3,3-Hexafluoro-2-propanol; rt

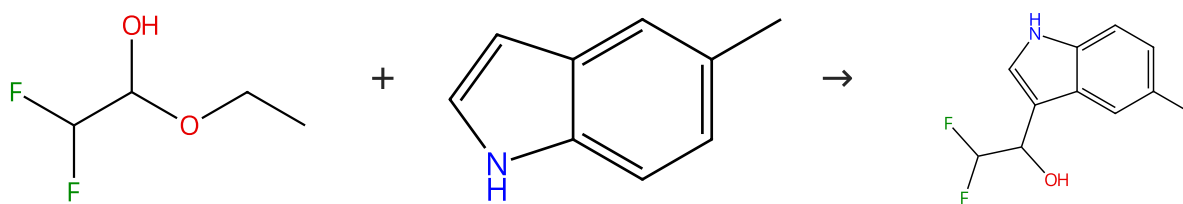
By: Gao, Juanjuan; et al

Molecules (2023), 28(22), 7522.

Experimental Protocols

Scheme 9 (1 Reaction)

Steps: 1 Yield: 95%



Suppliers (62)

Suppliers (86)

31-614-CAS-36085900

Steps: 1 Yield: 95%

Synthesis of Difluoromethylated Carbinols via a HFIP-Promoted Hydroxydifluoromethylation of Aniline, Indole, and Pyrrole Derivatives with Difluoroacetaldehyde Ethyl Hemiacetal

1.1 Reagents: 1,1,1,3,3,3-Hexafluoro-2-propanol
Solvents: Dichloromethane; 48 h, rt

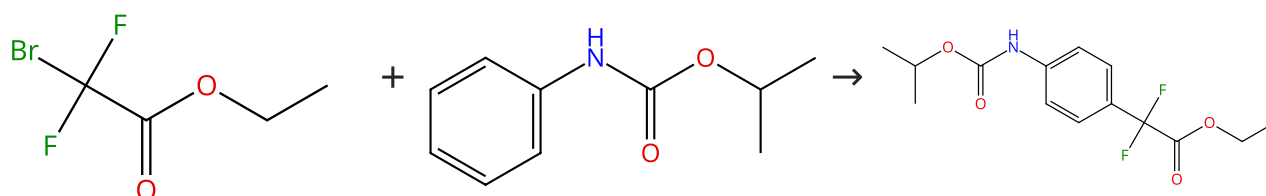
Experimental Protocols

By: Yang, Jianguo; et al

Journal of Organic Chemistry (2023), 88(7), 4790-4798.

Scheme 10 (1 Reaction)

Steps: 1 Yield: 95%



Suppliers (78)

Suppliers (42)

31-614-CAS-40564669

Steps: 1 Yield: 95%

Direct remote Csp²-H transformation of aromatic amines enabled by organophotoredox catalysis

1.1 Reagents: Cesium carbonate
Catalysts: 2,4,5,6-Tetra(9H-carbazol-9-yl)isophthalonitrile
Solvents: 1,4-Dioxane, 2,2-Dimethylpropanenitrile; 24 h, rt

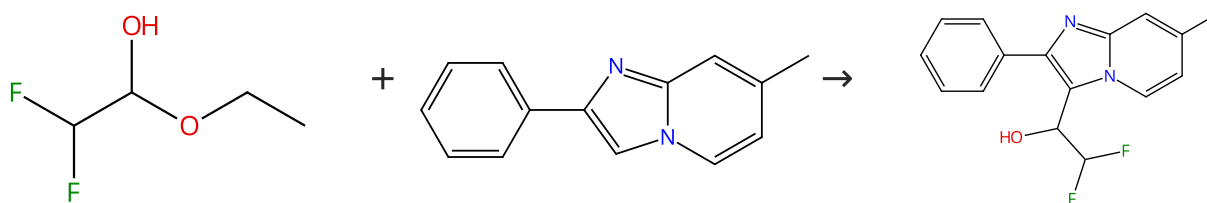
Experimental Protocols

By: Gou, Quan; et al

Green Chemistry (2024), 26(12), 7007-7012.

Scheme 11 (1 Reaction)

Steps: 1 Yield: 95%



Suppliers (62)

Suppliers (59)

31-614-CAS-38724275

Steps: 1 Yield: 95%

1,1,1,3,3,3-Hexafluoro-2-Propanol-Promoted Friedel-Crafts Reaction: Metal-Free Synthesis of C3-Difluoromethyl Carbinol-Containing Imidazo[1,2-a]pyridines at Room Temperature

1.1 Solvents: 1,1,1,3,3,3-Hexafluoro-2-propanol; 12 h, rt
1.2 Solvents: 1,1,1,3,3,3-Hexafluoro-2-propanol; rt

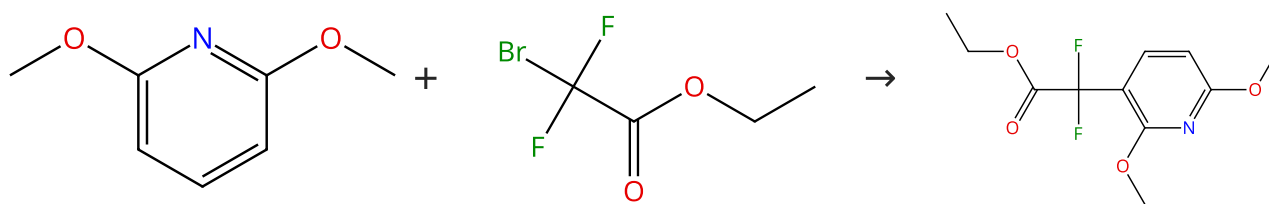
Experimental Protocols

By: Gao, Juanjuan; et al

Molecules (2023), 28(22), 7522.

Scheme 12 (1 Reaction)

Steps: 1 Yield: 95%



Suppliers (74)

Suppliers (78)

Suppliers (9)

31-614-CAS-41194895

Steps: 1 Yield: 95%

Ligand-assisted manganese-enabled direct C-H difluoromethylation of arenes

1.1 **Reagents:** Potassium carbonate
Catalysts: Dimanganese decacarbonyl, 2'-(Dicyclohexylphosphino)-*N,N*-dimethyl[1,1'-biphenyl]-2-amine
Solvents: Dichloromethane; 36 h, 40 °C

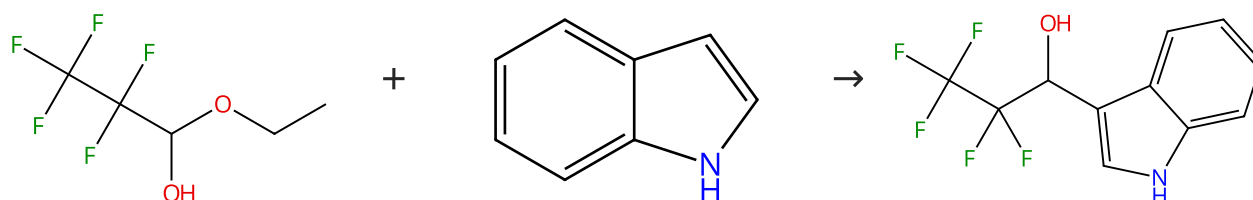
By: Zhou, Kehan; et al

Organic Chemistry Frontiers (2024), 11(17), 4874-4881.

Experimental Protocols

Scheme 13 (1 Reaction)

Steps: 1 Yield: 95%



Suppliers (13)

Suppliers (113)

Suppliers (2)

31-614-CAS-36085912

Steps: 1 Yield: 95%

Synthesis of Difluoromethylated Carbinols via a HFIP-Promoted Hydroxydifluoromethylation of Aniline, Indole, and Pyrrole Derivatives with Difluoroacetaldehyde Ethyl Hemiacetal

1.1 **Reagents:** 1,1,1,3,3,3-Hexafluoro-2-propanol
Solvents: Dichloromethane; 48 h, rt

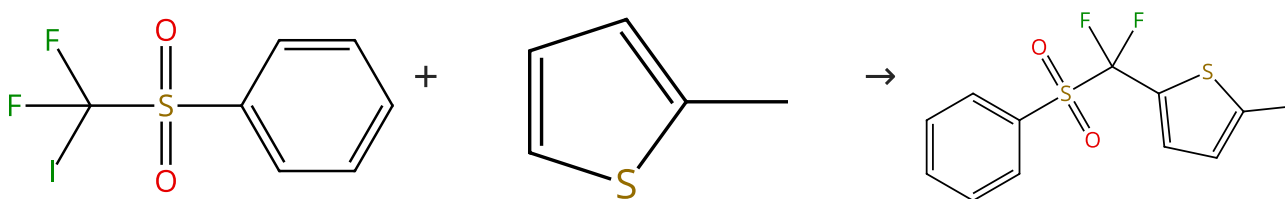
By: Yang, Jianguo; et al

Journal of Organic Chemistry (2023), 88(7), 4790-4798.

Experimental Protocols

Scheme 14 (1 Reaction)

Steps: 1 Yield: 94%



Suppliers (51)

Suppliers (71)

31-614-CAS-44133243

Steps: 1 Yield: 94%

Pd-catalyzed C-H(phenylsulfonyl)difluoromethylation of electron-rich heteroarenes

1.1 **Reagents:** Cesium carbonate
Catalysts: Tris(dibenzylideneacetone)dipalladium, 1,1'-(9,9-Dimethyl-9*H*-xanthene-4,5-diyl)bis[1,1-diphenylphosphine]
Solvents: Chloroform; 10 min, rt; 24 h, 60 °C

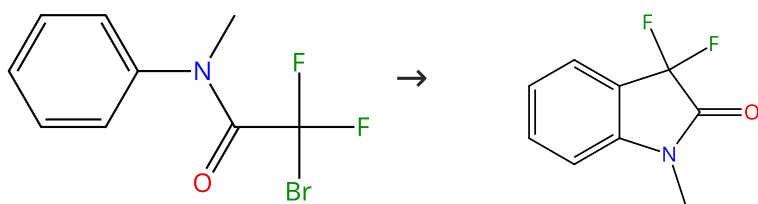
By: Liu, Zi-En; et al

Tetrahedron Letters (2025), 158, 155486.

Experimental Protocols

Scheme 15 (1 Reaction)

Steps: 1 Yield: 94%



Suppliers (16)

Suppliers (5)

31-614-CAS-32071019

Steps: 1 Yield: 94%

Divergent Regioselective Csp²-H Difluoromethylation of Aromatic Amines Enabled by Nickel Catalysis

By: Gou, Quan; et al

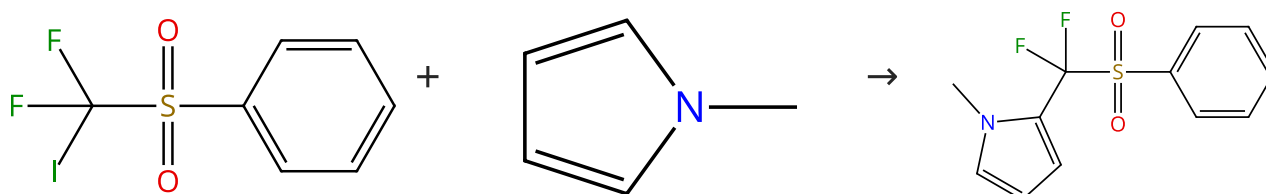
Organic Letters (2022), 24(19), 3549-3554.

- 1.1 **Reagents:** Sodium carbonate
Catalysts: 1,3-Bis(diphenylphosphino)propane, Methanesulfonic acid, 1,1,1-trifluoro-, nickel(2+) salt (2:1)
Solvents: 1,2-Dichloroethane; rt; 24 h, 133 °C

Experimental Protocols

Scheme 16 (1 Reaction)

Steps: 1 Yield: 94%



Suppliers (51)

Suppliers (75)

Suppliers (2)

31-614-CAS-44133237

Steps: 1 Yield: 94%

Pd-catalyzed C-H(phenylsulfonyl)difluoromethylation of electron-rich heteroarenes

By: Liu, Zi-En; et al

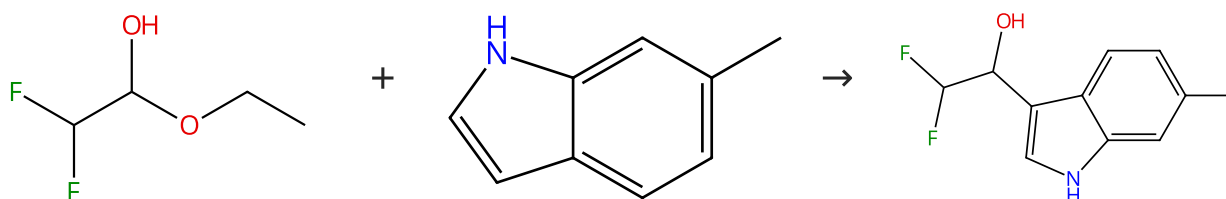
Tetrahedron Letters (2025), 158, 155486.

- 1.1 **Reagents:** Cesium carbonate
Catalysts: Tris(dibenzylideneacetone)dipalladium, 1,1'-(9,9-Dimethyl-9H-xanthene-4,5-diyl)bis[1,1-diphenylphosphine]
Solvents: Chloroform; 10 min, rt; 24 h, 60 °C

Experimental Protocols

Scheme 17 (1 Reaction)

Steps: 1 Yield: 94%



Suppliers (62)

Suppliers (77)

31-614-CAS-36085897

Steps: 1 Yield: 94%

Synthesis of Difluoromethylated Carbinols via a HFIP-Promoted Hydroxydifluoromethylation of Aniline, Indole, and Pyrrole Derivatives with Difluoroacetaldehyde Ethyl Hemiacetal

By: Yang, Jianguo; et al

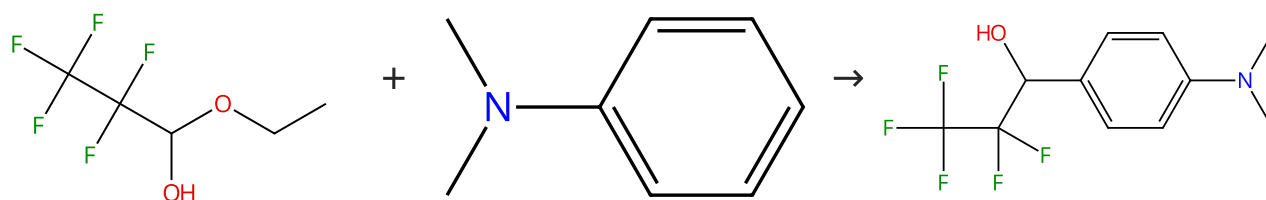
Journal of Organic Chemistry (2023), 88(7), 4790-4798.

- 1.1 **Reagents:** 1,1,1,3,3,3-Hexafluoro-2-propanol
Solvents: Dichloromethane; 48 h, rt

Experimental Protocols

Scheme 18 (1 Reaction)

Steps: 1 Yield: 93%



Suppliers (13)

Suppliers (90)

31-614-CAS-36085911

Steps: 1 Yield: 93%

1.1 Reagents: 1,1,1,3,3,3-Hexafluoro-2-propanol
Solvents: Dichloromethane; 48 h, rt

Experimental Protocols

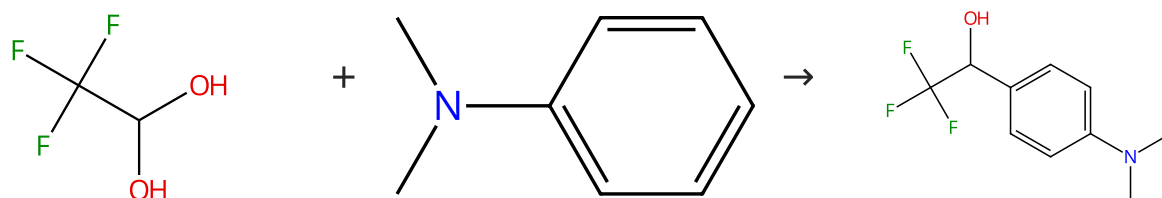
Synthesis of Difluoromethylated Carbinols via a HFIP-Promoted Hydroxydifluoromethylation of Aniline, Indole, and Pyrrole Derivatives with Difluoroacetaldehyde Ethyl Hemiacetal

By: Yang, Jianguo; et al

Journal of Organic Chemistry (2023), 88(7), 4790-4798.

Scheme 19 (1 Reaction)

Steps: 1 Yield: 93%



Suppliers (59)

Suppliers (90)

Suppliers (32)

31-614-CAS-36085908

Steps: 1 Yield: 93%

1.1 Reagents: 1,1,1,3,3,3-Hexafluoro-2-propanol
Solvents: Dichloromethane; 48 h, rt

Experimental Protocols

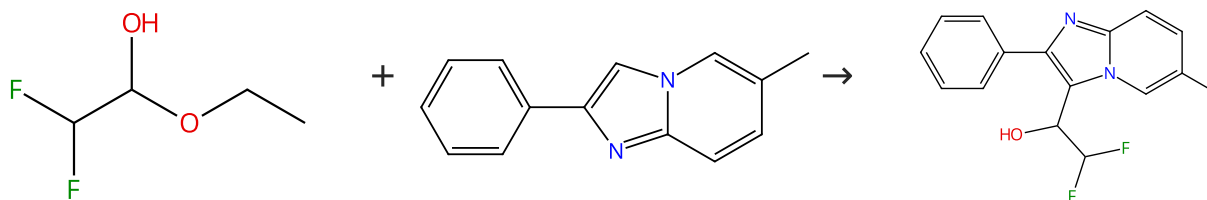
Synthesis of Difluoromethylated Carbinols via a HFIP-Promoted Hydroxydifluoromethylation of Aniline, Indole, and Pyrrole Derivatives with Difluoroacetaldehyde Ethyl Hemiacetal

By: Yang, Jianguo; et al

Journal of Organic Chemistry (2023), 88(7), 4790-4798.

Scheme 20 (1 Reaction)

Steps: 1 Yield: 93%



Suppliers (62)

Suppliers (54)

31-614-CAS-38724277

Steps: 1 Yield: 93%

1.1 Solvents: 1,1,1,3,3,3-Hexafluoro-2-propanol; 12 h, rt

1.2 Solvents: 1,1,1,3,3,3-Hexafluoro-2-propanol; rt

Experimental Protocols

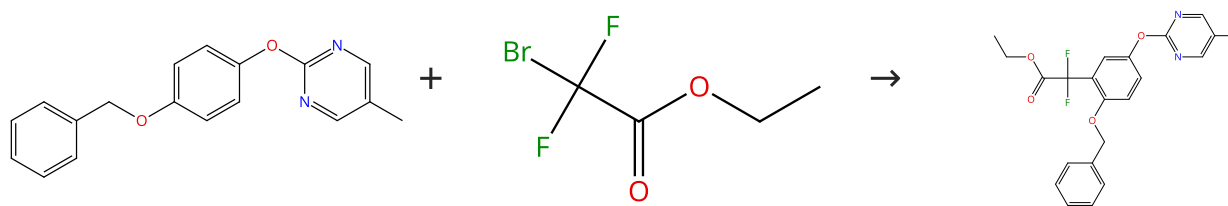
1,1,1,3,3,3-Hexafluoro-2-Propanol-Promoted Friedel-Crafts Reaction: Metal-Free Synthesis of C3-Difluoromethyl Carbinol-Containing Imidazo[1,2-a]pyridines at Room Temperature

By: Gao, Juanjuan; et al

Molecules (2023), 28(22), 7522.

Scheme 21 (1 Reaction)

Steps: 1 Yield: 93%



Suppliers (78)

31-085-CAS-23065219

Steps: 1 Yield: 93%

Ruthenium-Catalyzed Meta-Selective C-H Difluoromethylation of Phenol Derivatives1.1 **Reagents:** Sodium carbonate**Catalysts:** 2,4,6-Trimethylbenzoic acid, Bis(dichloro(η^6 -*p*-cymene)ruthenium), [1,1,1-Trifluoro-*N*-[(trifluoromethyl)sulfonyl- κO]methanesulfonamido- κO]silver**Solvents:** Toluene; 24 h, 150 °C

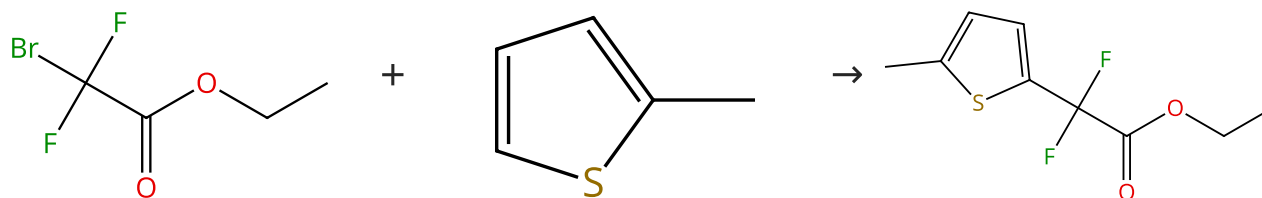
By: Xu, Xu; et al

Journal of Organic Chemistry (2020), 85(21), 13868-13876.

Experimental Protocols

Scheme 22 (1 Reaction)

Steps: 1 Yield: 93%



Suppliers (78)

Suppliers (71)

Suppliers (5)

31-614-CAS-41194898

Steps: 1 Yield: 93%

Ligand-assisted manganese-enabled direct C-H difluoromethylation of arenes1.1 **Reagents:** Potassium carbonate**Catalysts:** Dimanganese decacarbonyl, 2'-(Dicyclohexylphosphino)-*N,N*-dimethyl[1,1'-biphenyl]-2-amine**Solvents:** Dichloromethane; 36 h, 40 °C

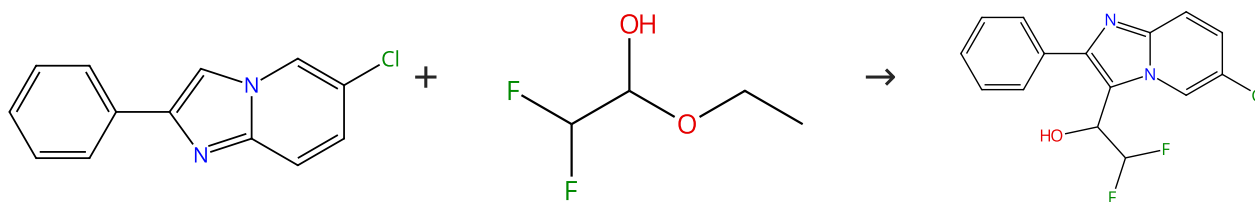
By: Zhou, Kehan; et al

Organic Chemistry Frontiers (2024), 11(17), 4874-4881.

Experimental Protocols

Scheme 23 (1 Reaction)

Steps: 1 Yield: 92%



Suppliers (57)

Suppliers (62)

31-614-CAS-38724281

Steps: 1 Yield: 92%

1,1,1,3,3,3-Hexafluoro-2-Propanol-Promoted Friedel-Crafts Reaction: Metal-Free Synthesis of C3-Difluoromethyl Carbinol-Containing Imidazo[1,2-a]pyridines at Room Temperature1.1 **Solvents:** 1,1,1,3,3,3-Hexafluoro-2-propanol; 12 h, rt1.2 **Solvents:** 1,1,1,3,3,3-Hexafluoro-2-propanol; rt

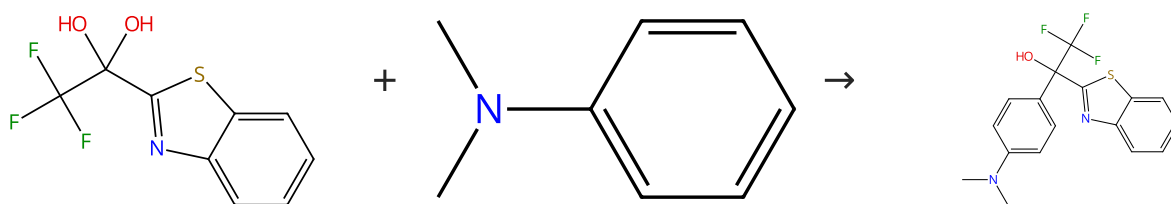
By: Gao, Juanjuan; et al

Molecules (2023), 28(22), 7522.

Experimental Protocols

Scheme 24 (1 Reaction)

Steps: 1 Yield: 92%



Suppliers (18)

Suppliers (90)

31-614-CAS-36085910

Steps: 1 Yield: 92%

Synthesis of Difluoromethylated Carbinols via a HFIP-Promoted Hydroxydifluoromethylation of Aniline, Indole, and Pyrrole Derivatives with Difluoroacetaldehyde Ethyl Hemiacetal

By: Yang, Jianguo; et al

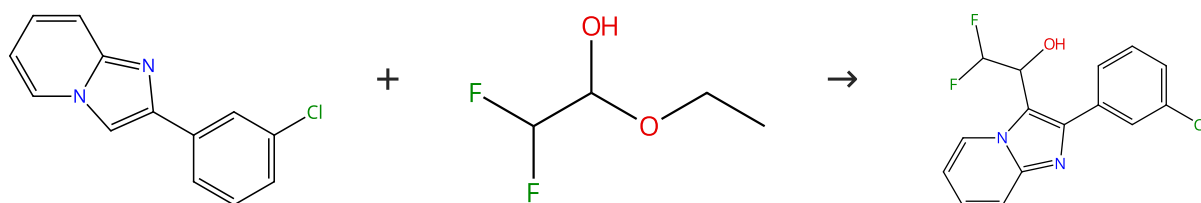
Journal of Organic Chemistry (2023), 88(7), 4790-4798.

1.1 Reagents: 1,1,1,3,3,3-Hexafluoro-2-propanol
Solvents: Dichloromethane; 48 h, rt

Experimental Protocols

Scheme 25 (1 Reaction)

Steps: 1 Yield: 92%



Suppliers (33)

Suppliers (62)

31-614-CAS-38724261

Steps: 1 Yield: 92%

1,1,1,3,3,3-Hexafluoro-2-Propanol-Promoted Friedel-Crafts Reaction: Metal-Free Synthesis of C3-Difluoromethyl Carbinol-Containing Imidazo[1,2-a]pyridines at Room Temperature

By: Gao, Juanjuan; et al

Molecules (2023), 28(22), 7522.

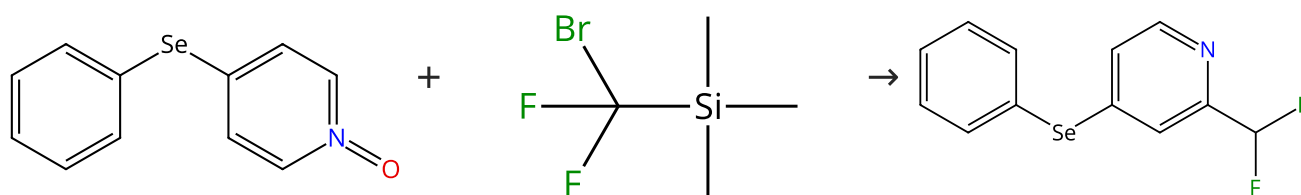
1.1 Solvents: 1,1,1,3,3,3-Hexafluoro-2-propanol; 12 h, rt

1.2 Solvents: 1,1,1,3,3,3-Hexafluoro-2-propanol; rt

Experimental Protocols

Scheme 26 (1 Reaction)

Steps: 1 Yield: 92%



Suppliers (56)

31-614-CAS-24002255

Steps: 1 Yield: 92%

Synthesis of Difluoroalkylated Heteroarenes via Difluoro carbene

By: Trifonov, Alexey L.; et al

Organic Letters (2021), 23(17), 6977-6981.

1.1 Reagents: Methyl triflate
Solvents: Dichloromethane; rt; 1 h, rt; rt → -20 °C

1.2 Reagents: Triphenylphosphine
Solvents: Acetonitrile; -30 °C; -30 °C → -42 °C

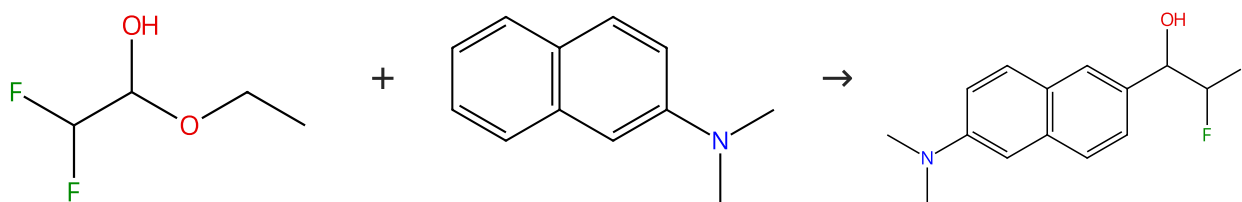
1.3 Reagents: Hexamethylphosphoramide; -42 °C; 2 h, -40 °C; 1 h, -40 °C → -16 °C; 15 min, rt; rt → -20 °C

1.4 Reagents: Triethylamine, Water; -20 °C; 12 h, rt

Experimental Protocols

Scheme 27 (1 Reaction)

Steps: 1 Yield: 92%



Suppliers (62)

Suppliers (62)

31-614-CAS-36085893

Steps: 1 Yield: 92%

Synthesis of Difluoromethylated Carbinols via a HFIP-Promoted Hydroxydifluoromethylation of Aniline, Indole, and Pyrrole Derivatives with Difluoroacetaldehyde Ethyl Hemiacetal

1.1 Reagents: 1,1,1,3,3,3-Hexafluoro-2-propanol
Solvents: Dichloromethane; 48 h, rt

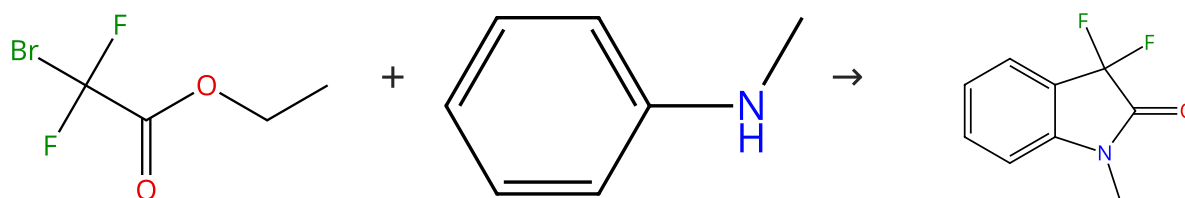
Experimental Protocols

By: Yang, Jianguo; et al

Journal of Organic Chemistry (2023), 88(7), 4790-4798.

Scheme 28 (1 Reaction)

Steps: 1 Yield: 92%



Suppliers (78)

Suppliers (73)

Suppliers (5)

31-614-CAS-32070938

Steps: 1 Yield: 92%

Divergent Regioselective Csp²-H Difluoromethylation of Aromatic Amines Enabled by Nickel Catalysis

1.1 Reagents: Sodium carbonate
Catalysts: 1,3-Bis(diphenylphosphino)propane, Methanesulfonic acid, 1,1,1-trifluoro-, nickel(2+) salt (2:1)
Solvents: 1,2-Dichloroethane; rt; 24 h, 133 °C

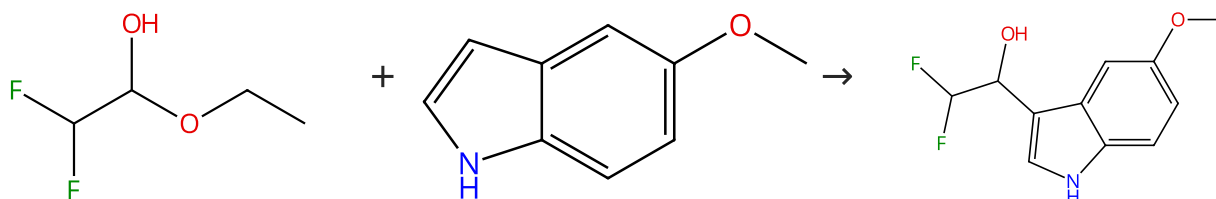
By: Gou, Quan; et al

Organic Letters (2022), 24(19), 3549-3554.

Experimental Protocols

Scheme 29 (1 Reaction)

Steps: 1 Yield: 92%



Suppliers (62)

Suppliers (104)

31-614-CAS-36085903

Steps: 1 Yield: 92%

Synthesis of Difluoromethylated Carbinols via a HFIP-Promoted Hydroxydifluoromethylation of Aniline, Indole, and Pyrrole Derivatives with Difluoroacetaldehyde Ethyl Hemiacetal

1.1 Reagents: 1,1,1,3,3,3-Hexafluoro-2-propanol
Solvents: Dichloromethane; 48 h, rt

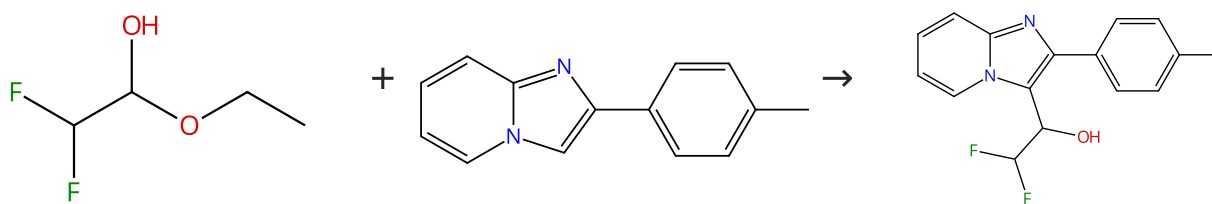
Experimental Protocols

By: Yang, Jianguo; et al

Journal of Organic Chemistry (2023), 88(7), 4790-4798.

Scheme 30 (1 Reaction)

Steps: 1 Yield: 92%



Suppliers (62)

Suppliers (53)

31-614-CAS-38724266

Steps: 1 Yield: 92%

1,1,1,3,3,3-Hexafluoro-2-Propanol-Promoted Friedel-Crafts Reaction: Metal-Free Synthesis of C3-Difluoromethyl Carbinol-Containing Imidazo[1,2-a]pyridines at Room Temperature

By: Gao, Juanjuan; et al

Molecules (2023), 28(22), 7522.

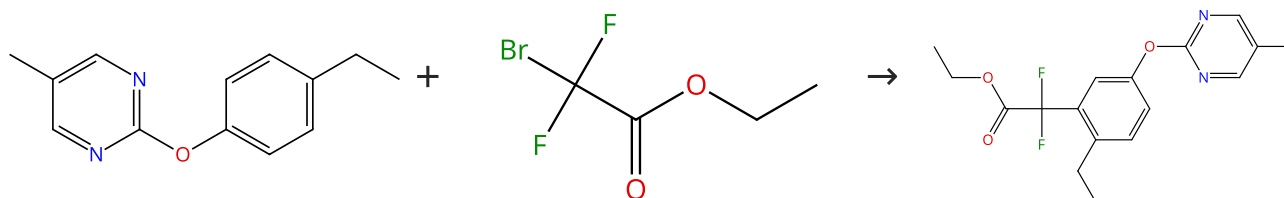
1.1 Solvents: 1,1,1,3,3,3-Hexafluoro-2-propanol; 12 h, rt

1.2 Solvents: 1,1,1,3,3,3-Hexafluoro-2-propanol; rt

Experimental Protocols

Scheme 31 (1 Reaction)

Steps: 1 Yield: 91%



Suppliers (78)

31-085-CAS-23068976

Steps: 1 Yield: 91%

Ruthenium-Catalyzed Meta-Selective C-H Difluoromethylation of Phenol Derivatives

By: Xu, Xu; et al

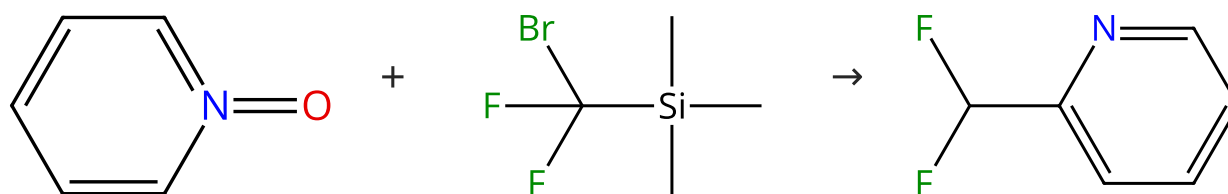
Journal of Organic Chemistry (2020), 85(21), 13868-13876.

1.1 Reagents: Sodium carbonate
Catalysts: 2,4,6-Trimethylbenzoic acid, Bis(dichloro(η⁶-p-cymene)ruthenium), [1,1,1-Trifluoro-N-[(trifluoromethyl)sulfonyl-κO]methanesulfonamido-κO]silver
Solvents: Toluene; 24 h, 150 °C

Experimental Protocols

Scheme 32 (1 Reaction)

Steps: 1 Yield: 91%



Suppliers (72)

Suppliers (56)

Suppliers (60)

31-614-CAS-24002249

Steps: 1 Yield: 91%

Synthesis of Difluoroalkylated Heteroarenes via Difluoro carbene

By: Trifonov, Alexey L.; et al

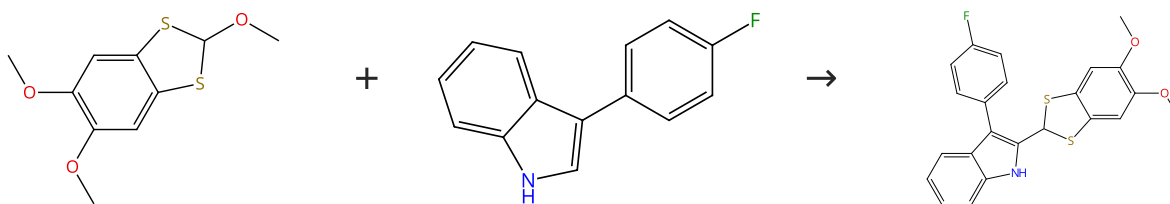
Organic Letters (2021), 23(17), 6977-6981.

1.1 Reagents: Methyl triflate
Solvents: Dichloromethane; rt; 1 h, rt; rt → -20 °C
1.2 Reagents: Triphenylphosphine
Solvents: Acetonitrile; -30 °C; -30 °C → -42 °C
1.3 Reagents: Hexamethylphosphoramide; -42 °C; 2 h, -40 °C; 1 h, -40 °C → -16 °C; 15 min, rt; rt → -20 °C
1.4 Reagents: Triethylamine, Water; -20 °C; 12 h, rt

Experimental Protocols

Scheme 33 (1 Reaction)

Steps: 1 Yield: 91%



Suppliers (54)

31-614-CAS-35206112

Steps: 1 Yield: 91%

Rapid ¹⁸F- and ¹⁹F-Difluoromethylation through Desulfurative Fluorination of Transient N-, O-, and C-Linked Dithioles

1.1 Solvents: Dichloromethane, 1,1,1,3,3,3-Hexafluoro-2-propanol; 3 h, rt

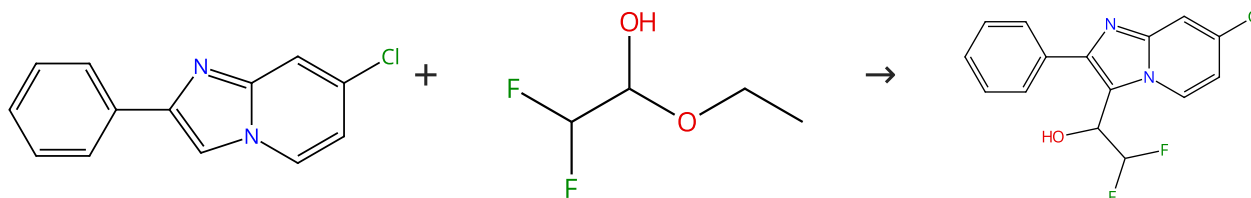
By: Newton, Josiah J.; et al

Experimental Protocols

Chemistry - A European Journal (2023), 29(5), e202202862.

Scheme 34 (1 Reaction)

Steps: 1 Yield: 91%



Suppliers (13)

Suppliers (62)

31-614-CAS-38724280

Steps: 1 Yield: 91%

1,1,1,3,3,3-Hexafluoro-2-Propanol-Promoted Friedel-Crafts Reaction: Metal-Free Synthesis of C3-Difluoromethyl Carbinol-Containing Imidazo[1,2-a]pyridines at Room Temperature

1.1 Solvents: 1,1,1,3,3,3-Hexafluoro-2-propanol; 12 h, rt

By: Gao, Juanjuan; et al

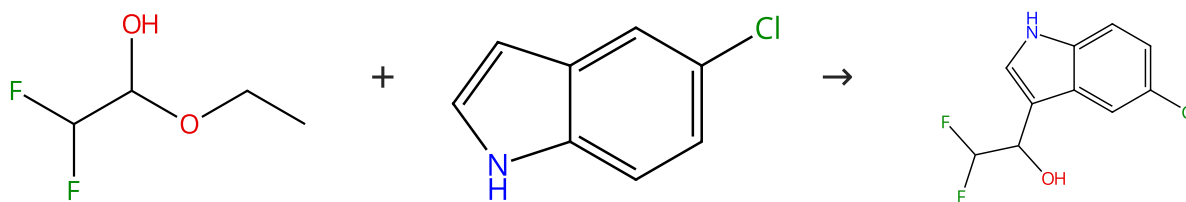
1.2 Solvents: 1,1,1,3,3,3-Hexafluoro-2-propanol; rt

Experimental Protocols

Molecules (2023), 28(22), 7522.

Scheme 35 (1 Reaction)

Steps: 1 Yield: 90%



Suppliers (62)

Suppliers (90)

31-614-CAS-36085899

Steps: 1 Yield: 90%

Synthesis of Difluoromethylated Carbinols via a HFIP-Promoted Hydroxydifluoromethylation of Aniline, Indole, and Pyrrole Derivatives with Difluoroacetaldehyde Ethyl Hemiacetal1.1 Reagents: 1,1,1,3,3,3-Hexafluoro-2-propanol
Solvents: Dichloromethane; 48 h, rt

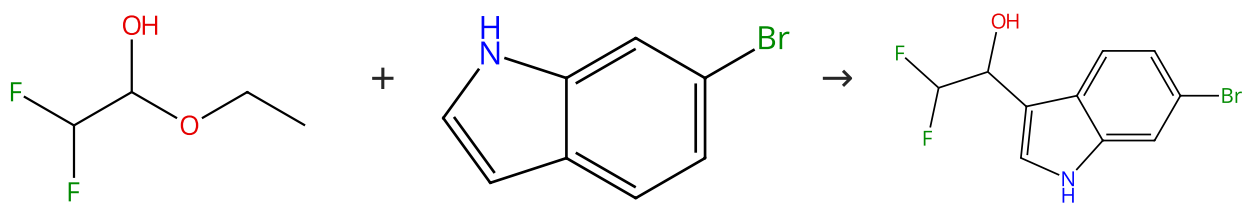
By: Yang, Jianguo; et al

Experimental Protocols

Journal of Organic Chemistry (2023), 88(7), 4790-4798.

Scheme 36 (1 Reaction)

Steps: 1 Yield: 90%



Suppliers (62)

Suppliers (94)

31-614-CAS-36085894

Steps: 1 Yield: 90%

Synthesis of Difluoromethylated Carbinols via a HFIP-Promoted Hydroxydifluoromethylation of Aniline, Indole, and Pyrrole Derivatives with Difluoroacetaldehyde Ethyl Hemiacetal

By: Yang, Jianguo; et al

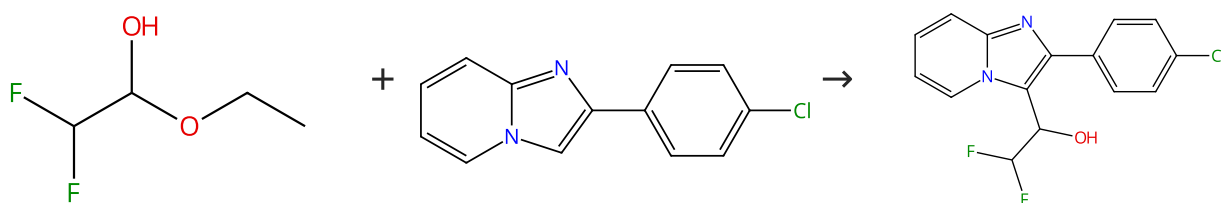
Journal of Organic Chemistry (2023), 88(7), 4790-4798.

1.1 Reagents: 1,1,1,3,3,3-Hexafluoro-2-propanol
Solvents: Dichloromethane; 48 h, rt

Experimental Protocols

Scheme 37 (1 Reaction)

Steps: 1 Yield: 90%



Suppliers (62)

Suppliers (60)

31-614-CAS-38724272

Steps: 1 Yield: 90%

1,1,1,3,3,3-Hexafluoro-2-Propanol-Promoted Friedel-Crafts Reaction: Metal-Free Synthesis of C3-Difluoromethyl Carbinol-Containing Imidazo[1,2-a]pyridines at Room Temperature

By: Gao, Juanjuan; et al

Molecules (2023), 28(22), 7522.

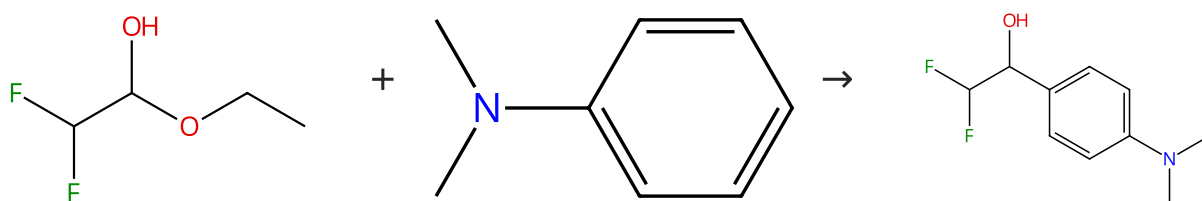
1.1 Solvents: 1,1,1,3,3,3-Hexafluoro-2-propanol; 12 h, rt

1.2 Solvents: 1,1,1,3,3,3-Hexafluoro-2-propanol; rt

Experimental Protocols

Scheme 38 (1 Reaction)

Steps: 1 Yield: 90%



Suppliers (62)

Suppliers (90)

Suppliers (10)

31-614-CAS-36085885

Steps: 1 Yield: 90%

Synthesis of Difluoromethylated Carbinols via a HFIP-Promoted Hydroxydifluoromethylation of Aniline, Indole, and Pyrrole Derivatives with Difluoroacetaldehyde Ethyl Hemiacetal

By: Yang, Jianguo; et al

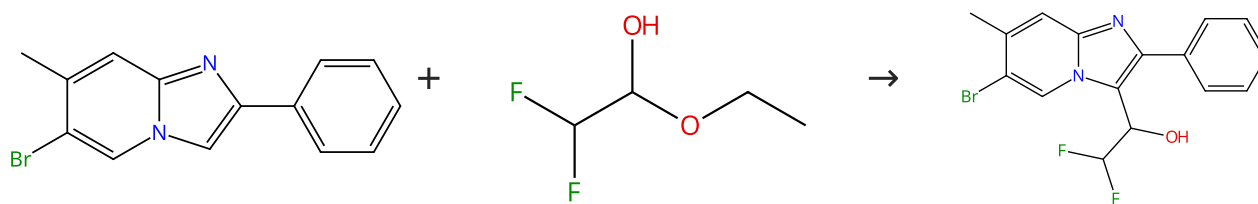
Journal of Organic Chemistry (2023), 88(7), 4790-4798.

1.1 Reagents: 1,1,1,3,3,3-Hexafluoro-2-propanol
Solvents: Dichloromethane; 48 h, rt

Experimental Protocols

Scheme 39 (1 Reaction)

Steps: 1 Yield: 90%



Suppliers (3)

Suppliers (62)

31-614-CAS-38724292

Steps: 1 Yield: 90%

1,1,1,3,3,3-Hexafluoro-2-Propanol-Promoted Friedel-Crafts Reaction: Metal-Free Synthesis of C3-Difluoromethyl Carbinol-Containing Imidazo[1,2-a]pyridines at Room Temperature

1.1 Solvents: 1,1,1,3,3,3-Hexafluoro-2-propanol; 12 h, rt

1.2 Solvents: 1,1,1,3,3,3-Hexafluoro-2-propanol; rt

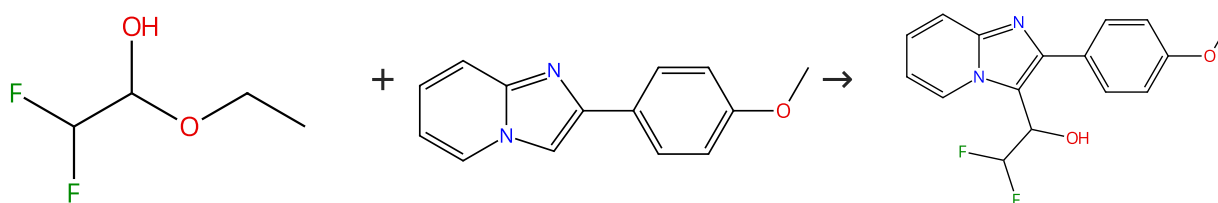
Experimental Protocols

By: Gao, Juanjuan; et al

Molecules (2023), 28(22), 7522.

Scheme 40 (1 Reaction)

Steps: 1 Yield: 90%



Suppliers (62)

Suppliers (44)

31-614-CAS-38724267

Steps: 1 Yield: 90%

1,1,1,3,3,3-Hexafluoro-2-Propanol-Promoted Friedel-Crafts Reaction: Metal-Free Synthesis of C3-Difluoromethyl Carbinol-Containing Imidazo[1,2-a]pyridines at Room Temperature

1.1 Solvents: 1,1,1,3,3,3-Hexafluoro-2-propanol; 12 h, rt

1.2 Solvents: 1,1,1,3,3,3-Hexafluoro-2-propanol; rt

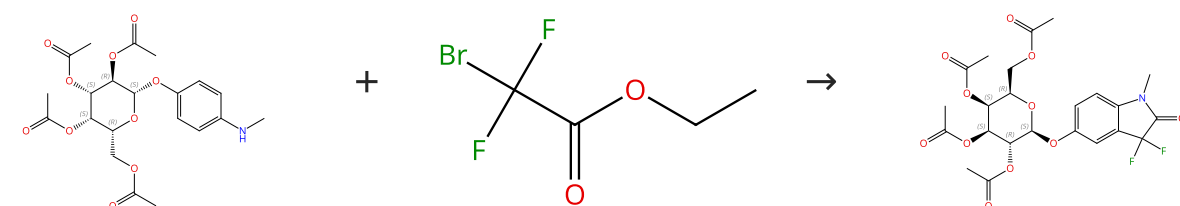
Experimental Protocols

By: Gao, Juanjuan; et al

Molecules (2023), 28(22), 7522.

Scheme 41 (1 Reaction)

Steps: 1 Yield: 90%



Absolute stereochemistry shown

Absolute stereochemistry shown

Suppliers (78)

31-614-CAS-32070949

Steps: 1 Yield: 90%

Divergent Regioselective Csp²-H Difluoromethylation of Aromatic Amines Enabled by Nickel Catalysis

1.1 Reagents: Sodium carbonate
Catalysts: 1,3-Bis(diphenylphosphino)propane, Methanesulfonic acid, 1,1,1-trifluoro-, nickel(2+) salt (2:1)
Solvents: 1,2-Dichloroethane; rt; 24 h, 133 °C

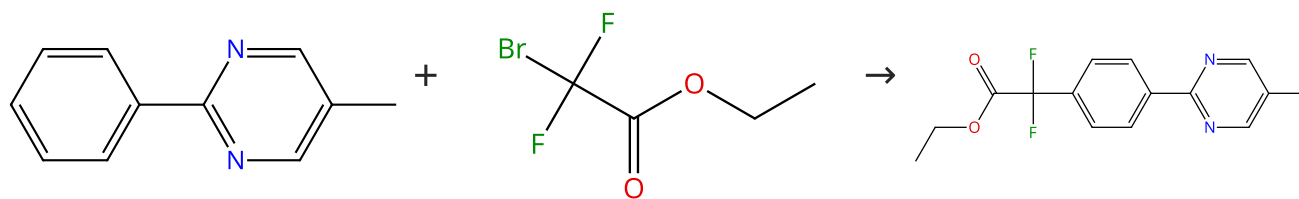
Experimental Protocols

By: Gou, Quan; et al

Organic Letters (2022), 24(19), 3549-3554.

Scheme 42 (1 Reaction)

Steps: 1 Yield: 90%



Suppliers (31)

Suppliers (78)

31-085-CAS-23107069

Steps: 1 Yield: 90%

Iron-Catalyzed Highly para-Selective Difluoromethylation of Arenes

1.1 **Reagents:** Potassium carbonate
Catalysts: Iron tetraphenylporphyrin chloride
Solvents: 1,2-Dichloroethane; 72 h, 140 °C

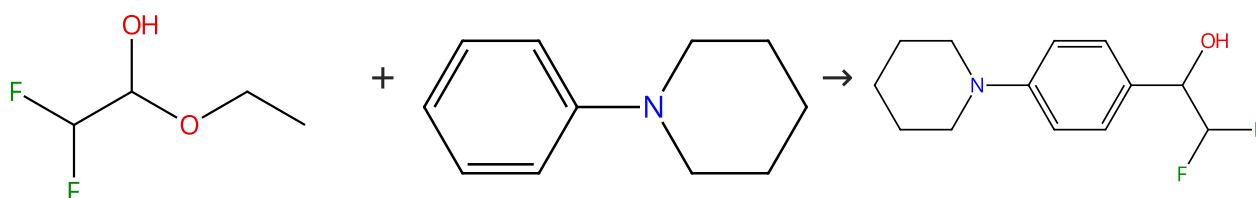
By: Fan, Wei-Tai; et al

Journal of the American Chemical Society (2020), 142(49), 20524-20530.

Experimental Protocols

Scheme 43 (1 Reaction)

Steps: 1 Yield: 89%



Suppliers (62)

Suppliers (55)

31-614-CAS-36085890

Steps: 1 Yield: 89%

Synthesis of Difluoromethylated Carbinols via a HFIP-Promoted Hydroxydifluoromethylation of Aniline, Indole, and Pyrrole Derivatives with Difluoroacetaldehyde Ethyl Hemiacetal

1.1 **Reagents:** 1,1,1,3,3,3-Hexafluoro-2-propanol
Solvents: Dichloromethane; 48 h, rt

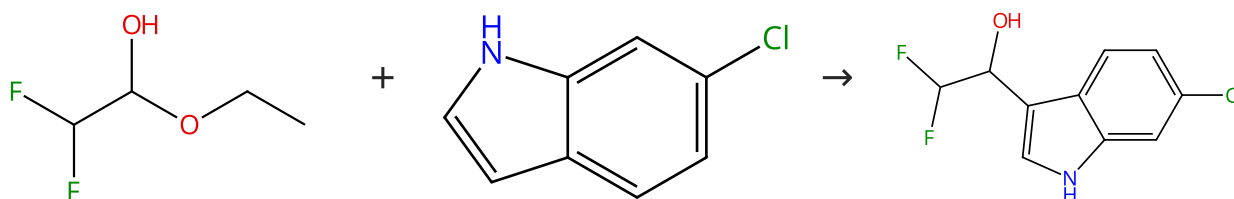
By: Yang, Jianguo; et al

Journal of Organic Chemistry (2023), 88(7), 4790-4798.

Experimental Protocols

Scheme 44 (1 Reaction)

Steps: 1 Yield: 89%



Suppliers (62)

Suppliers (90)

Suppliers (3)

31-614-CAS-36085904

Steps: 1 Yield: 89%

Synthesis of Difluoromethylated Carbinols via a HFIP-Promoted Hydroxydifluoromethylation of Aniline, Indole, and Pyrrole Derivatives with Difluoroacetaldehyde Ethyl Hemiacetal

1.1 **Reagents:** 1,1,1,3,3,3-Hexafluoro-2-propanol
Solvents: Dichloromethane; 48 h, rt

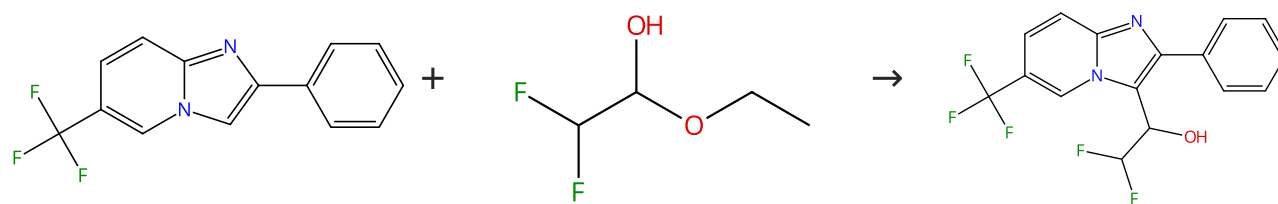
By: Yang, Jianguo; et al

Journal of Organic Chemistry (2023), 88(7), 4790-4798.

Experimental Protocols

Scheme 45 (1 Reaction)

Steps: 1 Yield: 89%



Suppliers (10)

Suppliers (62)

31-614-CAS-38724288

Steps: 1 Yield: 89%

1,1,1,3,3,3-Hexafluoro-2-Propanol-Promoted Friedel-Crafts Reaction: Metal-Free Synthesis of C3-Difluoromethyl Carbinol-Containing Imidazo[1,2-a]pyridines at Room Temperature

1.1 Solvents: 1,1,1,3,3,3-Hexafluoro-2-propanol; 12 h, rt

1.2 Solvents: 1,1,1,3,3,3-Hexafluoro-2-propanol; rt

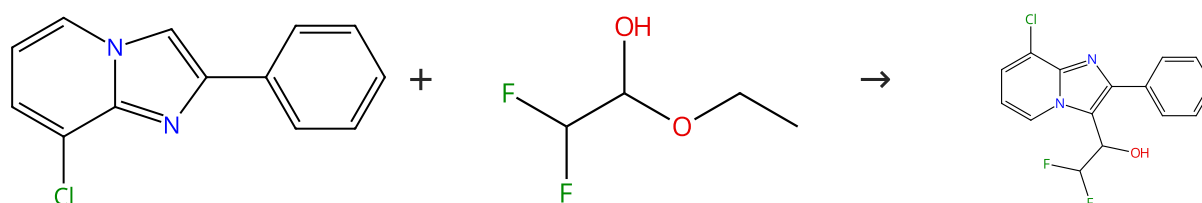
Experimental Protocols

By: Gao, Juanjuan; et al

Molecules (2023), 28(22), 7522.

Scheme 46 (1 Reaction)

Steps: 1 Yield: 89%



Suppliers (10)

Suppliers (62)

31-614-CAS-38724284

Steps: 1 Yield: 89%

1,1,1,3,3,3-Hexafluoro-2-Propanol-Promoted Friedel-Crafts Reaction: Metal-Free Synthesis of C3-Difluoromethyl Carbinol-Containing Imidazo[1,2-a]pyridines at Room Temperature

1.1 Solvents: 1,1,1,3,3,3-Hexafluoro-2-propanol; 12 h, rt

1.2 Solvents: 1,1,1,3,3,3-Hexafluoro-2-propanol; rt

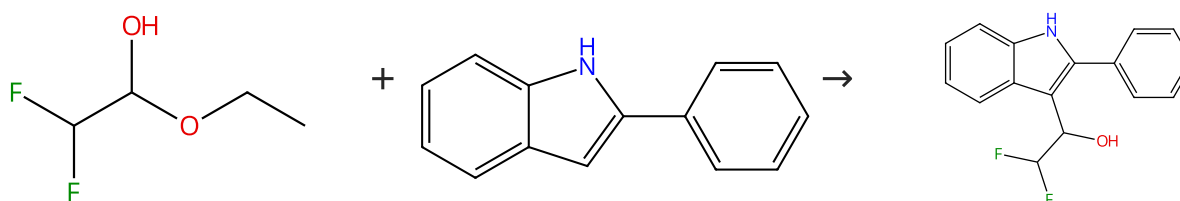
Experimental Protocols

By: Gao, Juanjuan; et al

Molecules (2023), 28(22), 7522.

Scheme 47 (1 Reaction)

Steps: 1 Yield: 89%



Suppliers (62)

Suppliers (85)

31-614-CAS-36085892

Steps: 1 Yield: 89%

Synthesis of Difluoromethylated Carbinols via a HFIP-Promoted Hydroxydifluoromethylation of Aniline, Indole, and Pyrrole Derivatives with Difluoroacetaldehyde Ethyl Hemiacetal

1.1 Reagents: 1,1,1,3,3,3-Hexafluoro-2-propanol
Solvents: Dichloromethane; 48 h, rt

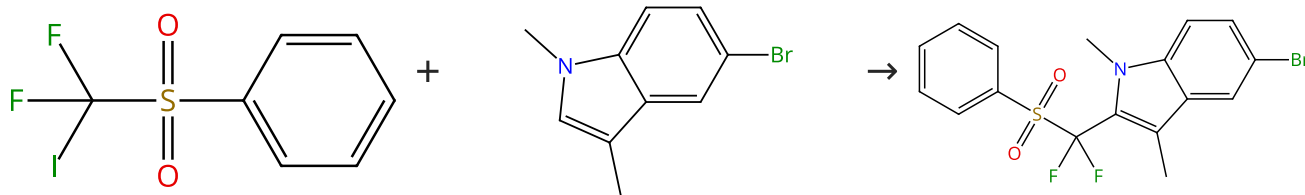
Experimental Protocols

By: Yang, Jianguo; et al

Journal of Organic Chemistry (2023), 88(7), 4790-4798.

Scheme 48 (1 Reaction)

Steps: 1 Yield: 88%



Suppliers (51)

Suppliers (53)

31-614-CAS-44133235

Steps: 1 Yield: 88%

Pd-catalyzed C-H(phenylsulfonyl)difluoromethylation of electron-rich heteroarenes

- 1.1 **Reagents:** Cesium carbonate
Catalysts: Tris(dibenzylideneacetone)dipalladium, 1,1'-(9,9-Dimethyl-9H-xanthene-4,5-diyl)bis[1,1-diphenylphosphine]
Solvents: Chloroform; 10 min, rt; 30 h, 40 °C

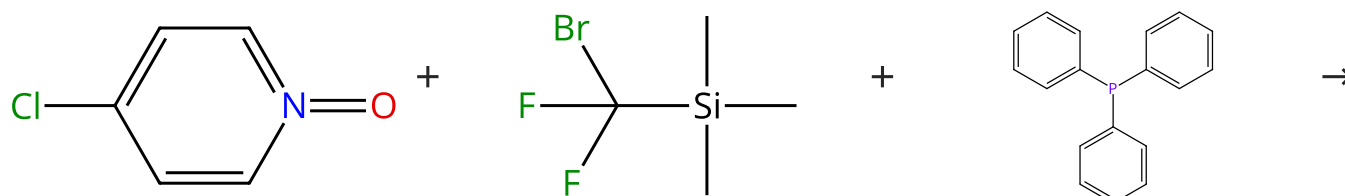
By: Liu, Zi-En; et al

Tetrahedron Letters (2025), 158, 155486.

Experimental Protocols

Scheme 49 (1 Reaction)

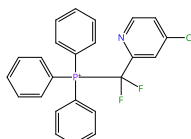
Steps: 1 Yield: 88%



Suppliers (70)

Suppliers (56)

Suppliers (87)

• Br⁻

31-614-CAS-24002276

Steps: 1 Yield: 88%

Synthesis of Difluoroalkylated Heteroarenes via Difluoro carbene

- 1.1 **Reagents:** Methyl triflate
Solvents: Dichloromethane; rt; 1 h, rt
 1.2 **Solvents:** Acetonitrile; -30 °C; -30 °C → -42 °C
 1.3 **Reagents:** Hexamethylphosphoramide; -42 °C; 2 h, -40 °C; 1 h, -40 °C → -16 °C; 15 min, rt

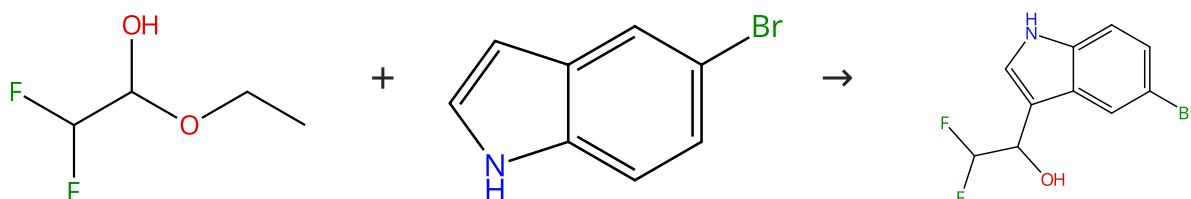
By: Trifonov, Alexey L.; et al

Organic Letters (2021), 23(17), 6977-6981.

Experimental Protocols

Scheme 50 (1 Reaction)

Steps: 1 Yield: 88%



Suppliers (62)

Suppliers (108)

31-614-CAS-36085896	Steps: 1 Yield: 88%	Synthesis of Difluoromethylated Carbinols via a HFIP-Promoted Hydroxydifluoromethylation of Aniline, Indole, and Pyrrole Derivatives with Difluoroacetaldehyde Ethyl Hemiacetal By: Yang, Jianguo; et al Journal of Organic Chemistry (2023), 88(7), 4790-4798.
1.1 Reagents: 1,1,1,3,3,3-Hexafluoro-2-propanol Solvents: Dichloromethane; 48 h, rt		
Experimental Protocols		